GOVERNMENT PURCHASES MULTIPLIER: EMPIRICAL RESULTS

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MOTIVATION AND DEFINITIONS

USUAL APPROACHES

- How to measure the multiplier?
- Ramey (2019) distinguishes three main approaches:
  1. aggregate country-level time series or panel estimates
  2. calibrated or estimated structural models
  3. subnational geographic cross-section or panel estimates
- Common problem — identification: estimating causal effects of government spending requires exogenous variation in policy.
- Instruments often have low correlation with the fiscal variable they are trying to explain.

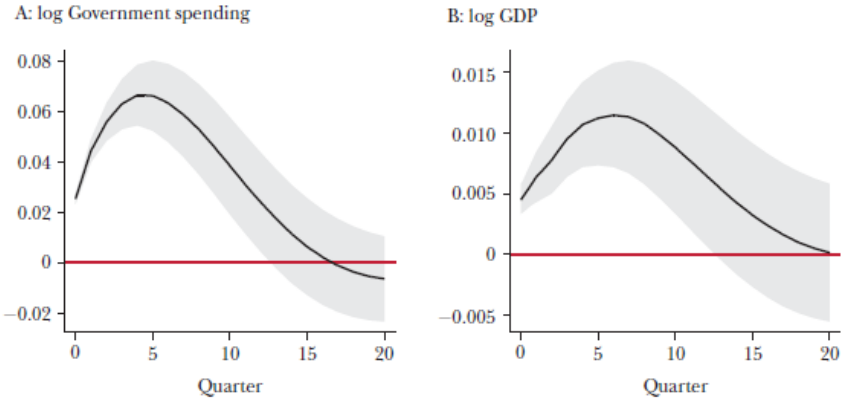
USUAL APPROACHES

- The third approach estimates across subnational units (states or provinces).
- These analyses have stronger identification due to more plausible assumptions and relevant instruments.
- This approach does not directly lead to macroeconomic estimates.
- Cross-sectional estimating equations include a constant term, netting out macroeconomic effects. Parameters estimated are only relative effects: *"If State A is awarded $1 more in defense prime contracts than the average state, by how much does its employment change relative to the average state?"*
- To infer national-level effects, researchers need a structural model or sufficient-statistics mapping.

WHAT IS "THE" MULTIPLIER?

- The effects of government spending on output are dynamic. Which ratio to use?
- Blanchard and Perotti (2002) calculate the ratio of output response at horizon *h* to the initial change of government spending.
- Mountford and Uhlig (2009) calculate the ratio of the cumulative response of output to the cumulative response of government spending, discounted.
- Another issue: regressions are often in logs, need to convert it to levels. What is the right ratio?

Estimated Impulse Response Functions for a Shock to Government Purchases

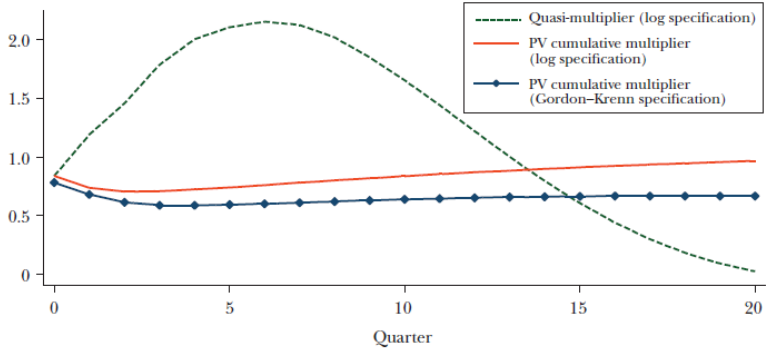


Source: Author.
Note: Estimated impulse responses based on structural vector autoregression (SVAR) estimates using quarterly data from 1939Q1– 2015Q4. The shaded area shows the 95-percent confidence bands. See the text and online appendix available with this paper at the journal website for more detail.

Estimated IRFs for a shock to government purchases. Ramey (2019)

Figure 2

Alternative Definitions of Multipliers: Multipliers by Horizon



Source: Author.

Note: The dotted and solid lines show multipliers calculated based on the log impulse responses shown in Figure 1. The dashed line shows the multiplier given by Blanchard-Perotti's (2002) method, which I call a *quasi-multiplier.* The solid line shows the the Mountford and Uhlig (2009) *present value (PV) cumulative multiplier.* The line with diamonds shows the PV cumulative multiplier using the impulse responses estimated using the Gordon–Krenn specification. See text and online appendix available with this paper at the journal website for more details.

Alternative multiplier definitions by horizon. Ramey (2019)

EMPIRICAL METHODS

VECTOR AUTOREGRESSIONS (VARS)

- A **VAR(p)** is a system of n equations where each variable is regressed on p lags of *all* variables:

$$Y_t = c + A_1 Y_{t-1} + A_2 Y_{t-2} + \cdots + A_p Y_{t-p} + U_t, \qquad U_t \sim (0, \Sigma_U)$$

  where Y_t is an $n \times 1$ vector (e.g. GDP, government spending, taxes) and each A_j is an $n \times n$ matrix of coefficients.
- Estimated equation by equation via OLS (Sims 1980). Consistent and efficient under stationarity.
- Impulse response functions (IRFs) trace out how each variable responds to a one-unit innovation in U_t, computed by iterating the system forward.
- **Problem**: the innovations U_t are *reduced-form* residuals — contemporaneously correlated. A shock to one equation is not "holding everything else fixed."

STRUCTURAL VARS (SVARS)

- A **structural VAR** recovers mutually uncorrelated *structural shocks* ε_t from the reduced-form innovations:

$$U_t = S\,\varepsilon_t, \qquad \mathbb{E}[\varepsilon_t \varepsilon_t'] = I$$

- The matrix S encodes contemporaneous causal relationships among variables.
- Since $\Sigma_U = SS'$, we have $n(n+1)/2$ restrictions from the data but n^2 unknowns in $S \Rightarrow$ need $n(n-1)/2$ additional *identifying restrictions*.
- Example: with $n = 3$ variables, the data gives 6 moment conditions but S has 9 free parameters $\Rightarrow$ need 3 restrictions.
- **Structural IRFs**: response of Y_{t+h} to a one-unit structural shock $\varepsilon_{j,t}$, given by column j of the matrices $\Phi_h S$, where Φ_h are the reduced-form MA coefficients.
- The choice of identifying restrictions is the central challenge in the fiscal multiplier literature.

LOCAL PROJECTIONS (JORDÀ 2005)

- Alternative to computing IRFs from a VAR: estimate a *separate regression* for each horizon $h = 0, 1, 2, \ldots, H$:

$$y_{t+h} = \mu_h + \beta_h x_t + \gamma_h' w_t + \xi_{h,t}$$

  where x_t is the shock of interest and w_t are controls (lags of Y_t).
- The sequence of coefficients $\{\beta_0, \beta_1, \ldots, \beta_H\}$ *is* the IRF directly — no need to iterate a system forward.
- Simple to combine with instrumental variables (LP-IV): replace x_t with an instrumented shock.

LOCAL PROJECTIONS VS. VARS

- **Advantages of LPs**:
  - Robust to VAR misspecification (lag length, omitted variables).
  - Easy to allow for nonlinearities and state dependence: interact x_t with a state variable (e.g. recession dummy).
- **Disadvantage**: less efficient in small samples — each horizon uses a separate regression, so estimates at long horizons are noisy.
- Key insight (Plagborg-Moller and Wolf 2021): VARs and LPs estimate the *same* IRFs in population; they differ only in finite-sample efficiency.

IDENTIFICATION

IDENTIFICATION STRATEGIES

1. **Short-run (zero) restrictions** (Cholesky/recursive):
   - Blanchard and Perotti (2002): government spending does not respond to GDP within the same quarter (institutional delays).
2. **Long-run restrictions** (Blanchard and Quah 1989):
   - Only supply shocks have permanent effects on output.
3. **Sign restrictions** (Uhlig 2005):
   - Impose signs on IRFs, not exact zeros. Weaker: yields *identified sets*.
4. **External instruments / proxy SVARs** (Stock and Watson 2012, Mertens and Ravn 2013):
   - Use an external z_t correlated with the structural shock but exogenous to others.
   - For fiscal policy: use narrative shocks as instruments.

WHY MILITARY SPENDING?

- The ideal experiment: randomly assign government spending across countries and time. Impossible in practice.
- **Military spending** is the best available substitute. Defense spending is driven primarily by geopolitical events (wars, threats), making it arguably more exogenous to the business cycle than other spending categories.
- Large, sudden changes: Korean War, Vietnam, post-9/11 buildup.
- Potential problems:
  - Other policies often change simultaneously (rationing, tax increases, price controls).
  - Timing: when does the economy respond — when spending is *announced* or when it is *spent*? Fiscal foresight matters.
- This motivates the distinction between different shock measures, each addressing these concerns differently.

HOW ARE FISCAL SHOCKS IDENTIFIED?

- **Barro and Redlick (2009)**: use the change in military spending $\Delta G_t^{\text{mil}}/Y_{t-1}$ directly as a regressor. Military spending driven by wars $\Rightarrow$ arguably exogenous to the business cycle.
- **Ramey (2011) military news**: read historical newspapers to identify dates when the public *first learned* about military buildups. Compute the expected present value of spending changes.
  - This *news* series leads actual spending — avoids the "fiscal foresight" problem.
- **Blanchard-Perotti (2002)**: residual from regressing government spending on lagged macro variables. Relies on spending not responding to GDP within the quarter.
- **Ben Zeev and Pappa (2017)**: use medium-horizon restrictions (Barsky and Sims 2011) — identify the shock orthogonal to current defense spending that *best explains future defense spending* over a 5-year horizon.
  - Statistical rather than narrative; higher multiplier estimates (≈ 1.4) but less precise.

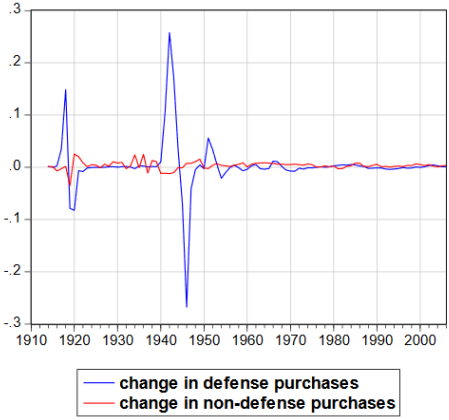


Figure 1

Changes in Defense and Non-Defense Government Purchases, 1914-2006

(expressed as ratios to the previous year's GDP)

Note: The figure shows the change in per capita real government purchases (nominal purchases divided by the GDP deflator), expressed as a ratio to the prior year's per capita real GDP. The blue graph is for defense purchases, and the red graph is for non-defense purchases by all levels of government. The government purchases data are from Bureau of Economic Analysis since 1929 and, before that, from Kendrick (1961). The GDP data are described at http://www.economics.harvard.edu/faculty/barro/data_sets_barro

Changes in defense and non-defense purchases as ratios to GDP. Barro and Redlick (2009)

THE BLANCHARD-PEROTTI (2002) SVAR

- Three-variable VAR: $Y_t = [\text{taxes, gov spending, GDP}]$, quarterly US data.
- Link VAR innovations U_t to structural shocks ε_t (Ramey 2016):

$$U_{1t} = b_{12}\, U_{2t} + b_{13}\, U_{3t} + \varepsilon_{1t} \quad \text{(taxes)}$$

$$U_{2t} = b_{21}\, U_{1t} + b_{23}\, U_{3t} + \varepsilon_{2t} \quad \text{(gov spending)}$$

$$U_{3t} = b_{31}\, U_{1t} + b_{32}\, U_{2t} + \varepsilon_{3t} \quad \text{(GDP)}$$

- **BP restrictions**:
  - $b_{21} = b_{23} = 0$: gov spending does not respond to taxes or GDP within the quarter (institutional delays). Equivalent to ordering G first.
  - $b_{13} = 2.08$: tax elasticity to GDP, set from institutional tax code data.
- These three restrictions identify all shocks. Results: spending multiplier $\approx$ 0.6–0.8; tax multiplier ≈ -0.7 to -1.3.

METHODOLOGICAL PITFALLS

INSTRUMENT RELEVANCE: RAMEY AND ZUBAIRY (2018)

- Use LP-IV to estimate fiscal multipliers with US quarterly data spanning 1889–2015.
- Two candidate instruments for government spending shocks:
  1. **Military news** (Ramey 2011): exogenous to the economy. Strong instrument at medium/long horizons, weak at impact.
  2. **Blanchard-Perotti shock**: strong at short horizons, weaker at longer horizons.
- Using both instruments together improves relevance across all horizons.
- Instrument relevance *varies by horizon* — matters which shock you use.

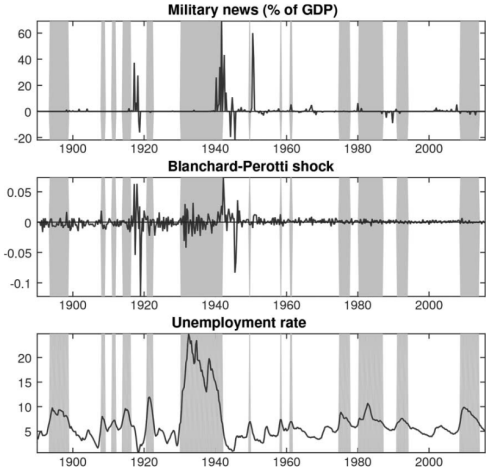


FIG. 3.—Military spending news, Blanchard-Perotti shock, and unemployment rate. Shaded areas indicate periods when the unemployment rate is above the threshold of 6.5 percent. Color version available as an online enhancement.

Military news, BP shocks, and unemployment. Ramey and Zubairy (2018)

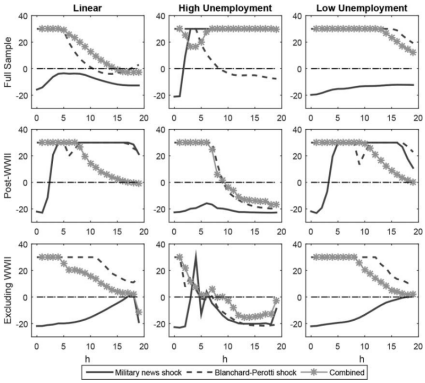


FIG. 4.—Tests of instrument relevance across states of slack. "Slack" is when the unemployment rate exceeds 6.5 percent. The lines show the difference between the effective F-statistic and the relevant threshold for the 5 percent level and are capped at 30. The effective F-statistics are from the regression of the sum of government spending through horizon h on the shock at t and all the other controls from the second stage, separately for the military news variable (solid line), the Blanchard-Perotti shock (dashed line), and both instruments (line with asterisks). The first column shows the linear case, the second column shows the high-unemployment state, and the last column shows the low-unemployment state. The full sample is 1890q1–2015q4, and the post-WWII sample spans 1947q3–2015q4. Color version available as an online enhancement.

Effective F-statistics: military news vs. BP instruments across horizons. Ramey and Zubairy (2018)

INSTRUMENT RELEVANCE AND PITFALLS

- **How to assess instrument relevance?** In IV, the first-stage regression projects the endogenous variable (government spending) on the instrument z_t.
- The **effective F-statistic** tests whether z_t has significant explanatory power. Rule of thumb (Stock and Yogo 2005):
  - $F > 10$: instrument is likely strong enough for reliable inference.
  - $F < 10$: *weak instrument* problem — IV estimates become biased and confidence intervals unreliable.
  - In LP-IV settings (horizon-specific), effective F-statistics and appropriate critical values are needed (Montiel Olea and Pflueger 2013).
- Military news: F high at medium/long horizons, low at impact.
  BP shock: F high at short horizons, low at longer horizons.
  Both together: relevance substantially improved across all horizons.

LOGS VS. LEVELS: THE GORDON-KRENN SPECIFICATION

- VARs are often estimated in **logs**. The IRF gives elasticities (percentage responses). To convert to a dollar multiplier, multiply by $\bar{Y}/\bar{G}$.
- **Problem**: Y/G varies enormously over time — from 2 (WW2) to 24 in US historical data. Using a sample average $\bar{Y}/\bar{G}$ as conversion factor biases the multiplier *upward*.
- Worse: Y/G is *procyclical* $\Rightarrow$ using a fixed conversion factor makes multipliers appear more countercyclical (larger in recessions) than they really are.
- **Gordon-Krenn (2010) solution**: divide all variables by trend (potential) GDP: $\tilde{X}_t = X_t/Y_t^*$
  - Puts spending and output in the same units.
  - Multiplier estimated *directly* from IRF coefficients — no conversion factor needed.
- Impact is dramatic: the BP quasi-multiplier (log specification) peaks at ≈ 2.2, while the PV cumulative multiplier using Gordon-Krenn gives ≈ 0.6. Both specification *and* multiplier definition matter.

AGGREGATE RESULTS

COMPARING IDENTIFICATION METHODS (RAMEY 2016)

- Ramey (2016) provides a systematic comparison of three identification methods applied to the same dataset:
  1. Blanchard-Perotti (Cholesky ordering)
  2. Ramey military news (narrative)
  3. Ben Zeev-Pappa news (medium-horizon restrictions)
- Key questions: Do different methods give the same IRFs? Are the instruments equally strong? Do the multipliers agree?
- Spoiler: the methods *disagree* on the short-run dynamics (especially consumption and wages), but BP and Ramey news converge on a multiplier of roughly 0.6–1.0; BZP gives higher estimates ($\approx$ 1.1–1.4) but with large standard errors.

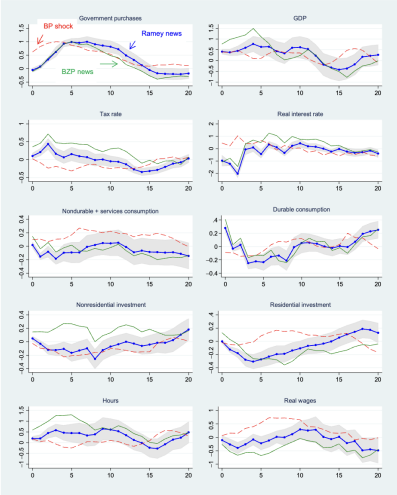


Fig. 5 Comparison of the effects of government spending shocks (*BP*: Blanchard–Perotti; *BZP*: Ben Zeev–Pappa). Light gray bands are 90% confidence bands.

IRFs to a government spending shock (post-WWII). BP, Ramey news, BZP news. Ramey (2016)

Table 4 Multiplier estimates (HAC standard errors in parenthesis)

Horizon (in quarters)	Blanchard–Perotti	Ramey news	Ben Zeev–Pappa news
0	0.65 (0.24)	−7.53 (7.26)	−7.37 (5.85)
4	0.37 (0.23)	1.37 (0.33)	2.91 (1.13)
8	0.39 (0.32)	0.80 (0.25)	1.41 (0.61)
12	0.39 (0.44)	0.77 (0.27)	1.24 (0.71)
16	0.40 (0.58)	0.60 (0.36)	1.10 (1.01)
20	0.44 (0.63)	0.69 (0.48)	1.17 (1.46)

Notes: Multipliers estimated using Eq. (19). All regressions also include two lags of the shock (to mop up any serial correlation), real GDP (divided by potential GDP), real government purchases (divided by potential GDP), the tax rate, and a quadratic trend.

LP-IV cumulative multiplier estimates (Gordon-Krenn specification, $\sum_{i=0}^{h} \Delta Y_{t+i} / \sum_{i=0}^{h} \Delta G_{t+i}$) by horizon and method. Ramey (2016)

WHAT DO DIFFERENT METHODS GIVE US?

- Ramey (2019) surveys the literature and collects multiplier estimates across methods and samples.
- **Key finding**: most aggregate time-series estimates cluster between **0.6 and 0.8** — well below 1.
- Ben Zeev and Pappa (2017) are the outlier with estimates of 1.1–2.0, but with large standard errors.
- Estimated NK DSGE models give similar numbers: **0.6 to 1.4**, depending on the model.
- The range narrows considerably once researchers use consistent methodology (Gordon-Krenn specification, proper instruments, levels instead of logs).

Table 1

Estimates of Government Spending Multipliers Using Aggregate Data, No State Dependence

(almost all are cumulative multipliers, typically over horizons between 0 to 20 quarters)

Method/Sample	*Multipliers*	*Comments*
A: Time series analysis		
Updated implementation of Blanchard and Perotti (2002) identified SVAR		The tax response is positive for the 1939Q1–2015Q4 period, but is essentially 0 for the later periods.
1939Q1–2015Q4	0.6 to 0.8	
1947Q1–2015Q4	0.6 to 0.7	
Military news shocks, local projections		Tax response is positive for 1939Q1–2015Q4 period.
Ramey and Zubairy (2018) military news		
1889Q1–2015Q4	0.6 to 0.8	SE from 0.04 to 0.06
1939Q1–2015Q4	0.7 to 0.8	SE from 0.05 to 0.1
1947Q1–2015Q4	0.5 to 0.7	SE from 0.15 to 0.2
Ben Zeev and Pappa (2017) news, 1947Q1–2007Q4[a]	1.1 to 2	SE from 0.6 to 1
Hall (2019), Barro and Redlick (2011)—based on regressions using annual defense spending.	0.6 to 0.7	The Barro–Redlick analysis nets out effects of changes in tax rates.
Mountford and Uhlig (2009), SVAR with sign restrictions	0.65	Deficit-financed increase in government spending.
Iltzetzki, Mendoza, and Végh (2013), Blanchard–Perotti identification in SVAR, quarterly data, 1960–2007, 44 countries high-income countries	0.3 to 0.7	
Corsetti, Meier, and Müller (2012)	0.7	Based on unconditional model results reported in their Figure 1.
Leigh et al. (2010), Guajardo, Leigh, and Pescatori (2014), 17 OECD countries, 1978–2009, narrative method identification of spending-based fiscal consolidations	0.3	
Alesina, Favero, and Giavazzi (forthcoming). Narrative analysis of austerity plans, 16 OECD economies from 1978–2014.	0.3	

Multiplier estimates from aggregate time series. Ramey (2019), Table 1A

Table 1

Estimates of Government Spending Multipliers Using Aggregate Data, No State Dependence

(almost all are cumulative multipliers, typically over horizons between 0 to 20 quarters)

Method/Sample	*Multipliers*	*Comments*
B: Estimated New Keynesian DSGE models		
Cogan et al. (2010), estimated Smets–Wouters DSGE model on US data	0.6 to 0.7	Based on my visual inspection of figures 2, 3, and 4.
Coenen et al. (2012), large-scale macro models used by central banks and IMF, United States and Europe	0.7 to 1	Based on the two-year cumulative multipliers shown in the upper left graph in figure 6.
Zubairy (2014), estimated medium-scale DSGE model on US data	0.7 to 1.05	Deficit financed, model features deep habits.
Leeper, Traum, and Walker (2017), estimated DSGE model on US data	0.7 to 1.36	Active monetary policy, table 7
Sims and Wolff (2018a)	1.07	The multiplier above 1 is due to estimated complementarity of government spending with private consumption.

Multiplier estimates from estimated NK DSGE models. Ramey (2019), Table 1B

THE 2009 STIMULUS: ARRA

- The American Recovery and Reinvestment Act (ARRA, 2009) was the largest peacetime fiscal expansion in US history at the time: $787 billion (later revised to $831 billion). Eclipsed only by the 2020 pandemic packages.
- It provided a natural experiment for estimating the multiplier, using both aggregate and subnational variation.
- Ramey (2019) collects estimates from many studies:
  - Model-based (DSGE) estimates: 0.5–1.8 depending on assumptions about monetary policy accommodation.
  - Cross-sectional estimates: typically larger, but need to be converted to national equivalents.

Table 3

Multipliers for the American Recovery and Reinvestment Act (ARRA)

Method/Sample	*Peak cumulative multipliers within first 5 years*	*Comments*
Cogan et al. (2010)	0.6 to 0.7	
Coenen et al. (2012), large-scale macro models used by central banks and IMF, US, and Europe		From figure 7. These are the peak instantaneous multipliers.
No monetary accommodation	0.3 to 0.5	
1 year monetary accommodation	0.4 to 0.6	
2 years monetary accommodation	0.5 to 1.8	
Drautzburg and Uhlig (2015), medium-scale New Keynesian DSGE model, with ZLB, credit constraints	0.5	Multipliers become negative in the long run because of the necessary increase in taxation.
Chodorow-Reich (forthcoming), based on cross-state estimates and theoretical arguments about the relationship between subnational and national multipliers at the ZLB.		
Gross State Product multiplier	1.7 to 2	
Cost per job year	2 job-years per $100K	

Multiplier estimates for ARRA. Ramey (2019)

Table 4

Conversion of Chodorow-Reich Estimates to Nationally Representative Estimates

	Cumulative employment multiplier estimates (number of job-years created per $100K of ARRA spending)
Chodorow-Reich (forthcoming) headline estimates (his table 1, column 4)	2.01 (0.59)
Weighted estimates (using December 2008 population of state)	1.15 (0.72)
Weighted estimates based on total spending, including induced spending by states	0.89 (0.45)

Note: ARRA is American Recovery and Reinvestment Act. Estimates presented in the last two rows are the author's estimates, based on Chodorow-Reich's programs and data in his forthcoming paper. See the text and online appendix for more detail and programs.

Converting subnational to national multiplier estimates. Ramey (2019)

STATE DEPENDENCE

STATE DEPENDENCE: RAMEY AND ZUBAIRY (2018)

- Does the multiplier depend on the *state of the economy*? This is one of the most debated questions in the post-crisis literature.
- Auerbach and Gorodnichenko (2012) argue yes: multiplier is larger in recessions.
- Ramey and Zubairy (2018) revisit using LP-IV with consistent methodology:
  - Multipliers 0.3–0.8 across specifications.
  - **No strong evidence of state dependence** once the specification is done correctly (levels, not logs; instrument relevance accounted for).
  - The log-to-level conversion using a time-varying Y/G ratio is a key source of bias in earlier work.
- What about the **ZLB**? RZ test using a near-zero interest rate indicator (covering both the 1930s–40s and 2008–15). Multiplier is somewhat higher (≈ 1.5), but estimates are imprecise due to few ZLB episodes.

SUBNATIONAL EVIDENCE

NAKAMURA AND STEINSSON (2014)

- Exploit regional variation in military procurement across US states.
- States differ in sensitivity to national military buildups: California receives $\approx 3\times$ the national average, Illinois $\approx 0.5\times$.
- **Bartik (shift-share) IV**: instrument state-level spending with $z_{s,t} = f_s \times g_t$, where f_s is the state's historical share of national military prime contracts and g_t is the change in national military spending per capita. The "shift" g_t is exogenous (driven by wars); the "share" f_s determines exposure.
- Time fixed effects net out all aggregate shocks (monetary policy, taxes, etc.).
- This identifies the *open economy relative multiplier*: the effect of a relative increase in spending on one region compared to others.

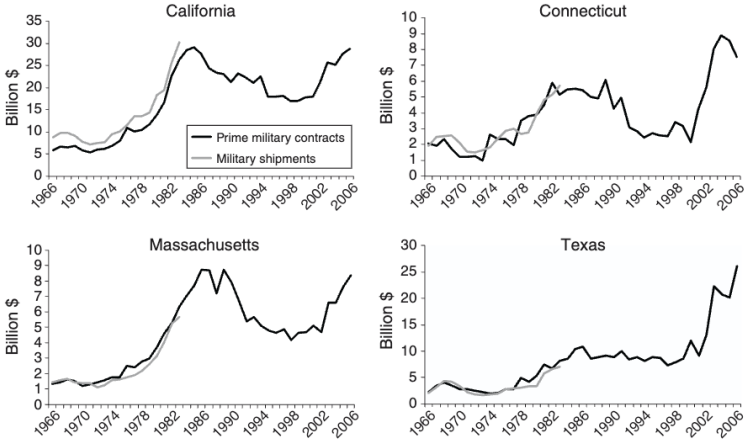


FIGURE 2. PRIME MILITARY CONTRACTS AND MILITARY SHIPMENTS

Prime military contracts by state. Nakamura and Steinsson (2014)

NAKAMURA AND STEINSSON (2014): RESULTS

- Open economy relative multiplier ≈ 1.5 (range 1.4–1.9).
- During high unemployment: multiplier rises to 3.5–4.3.
- **Interpretation**: the open economy relative multiplier ≠ the closed economy aggregate multiplier.
  - Equivalent to the multiplier in a *small open economy with a fixed exchange rate* (monetary union).
  - Captures demand spillovers within a region without aggregate monetary/fiscal offsets.

Table 2—The Effects of Military Spending

	Output		Output defl. state CPI		Employment		CPI	Population
	States	Regions	States	Regions	States	Regions	States	States
Prime military contracts	1.43	1.85	1.34	1.85	1.28	1.76	0.03	−0.12
	(0.36)	(0.58)	(0.36)	(0.71)	(0.29)	(0.62)	(0.18)	(0.17)
Prime contracts plus military compensation	1.62	1.62	1.36	1.44	1.39	1.51	0.19	0.07
	(0.40)	(0.84)	(0.39)	(0.96)	(0.32)	(0.91)	(0.16)	(0.21)
Observations	1,989	390	1,989	390	1,989	390	1,763	1,989

Notes: Each cell in the table reports results for a different regression with a shorthand for the main regressor of interest listed in the far left column. A shorthand for the dependent variable is stated at the top of each column. The dependent variable is a two-year change divided by the initial value in each case. Output and employment are per capita. The regressor is the two-year change divided by output. Military spending variables are per capita except in Population regression. Standard errors are in parentheses. All regressions include region and time fixed effects, and are estimated by two-stage least squares. The sample period is 1966–2006 for output, employment, and population, and 1969–2006 for the CPI. Output is state GDP, first deflated by the national CPI and then by our state CPI measures. Employment is from the BLS payroll survey. The CPI measure is described in the text. Standard errors are clustered by state or region.

Open economy relative multiplier estimates. Nakamura and Steinsson (2014)

WHY IS THE OPEN ECONOMY RELATIVE MULTIPLIER LARGE?

- A US state is like a *small open economy with a fixed exchange rate* (members of a currency union).
- When one state receives extra government spending:
  1. **No monetary offset**: the Fed does not raise interest rates in response to spending in one state $\Rightarrow$ no crowding out via higher rates.
  2. **No exchange rate appreciation**: the state cannot appreciate its currency $\Rightarrow$ no loss of competitiveness.
  3. **Demand stays local**: workers spend their income locally, amplifying the effect.
- At the *national* level, all three channels work against the multiplier: the central bank tightens, the exchange rate appreciates, and future taxes must rise.
- Therefore: open economy relative multiplier $>$ closed economy aggregate multiplier.
- The open economy relative multiplier is a powerful *test statistic*: NK model with GHH preferences predicts ≈ 1.5; neoclassical model only ≈ 0.4.

Table 7–Government Spending Multiplier in GHH Model

	Closed economy aggregate multiplier	Open economy relative multiplier
Panel A. Sticky prices		
Volcker-Greenspan monetary policy	0.12	1.42
Constant real rate	7.00	1.42
Constant nominal rate	∞	1.42
Constant nominal rate ($\rho_g = 0.50$)	8.73	2.04
Panel B. Flexible prices		
Constant income tax rates	0.00	0.30
Balanced budget	−0.18	0.30

Notes: The table reports the government spending multiplier for output deflated by the regional CPI for the model presented in the text with the GHH preferences specification. Panel A presents results for the model with sticky prices, while panel B presents results for the model with flexible prices. The first three rows differ only in the monetary policy being assumed. The fourth row varies the persistence of the government spending shock relative to the baseline parameter values. The fifth and sixth rows differ only in the tax policy being assumed.

Model-implied multipliers. Nakamura and Steinsson (2014)

FROM LOCAL TO AGGREGATE: THE KEY CHALLENGE

- Cross-sectional regressions include a constant term ⇒ all *aggregate* effects are netted out. The estimate is purely relative: "If State A gets $1 more than average, how much does its output rise relative to average?"
- **The relationship between local and aggregate multipliers depends on**:
  - Whether there are spillovers across regions (trade linkages).
  - Whether the aggregate economy is at the zero lower bound.
  - How the spending is financed (federal transfers vs. local taxes).
- There is *no general rule*: in some cases subnational multipliers overestimate the aggregate; in others they underestimate it.
- To go from local to aggregate, one needs a *structural macro model*.

DUPOR, KARABARBOUNIS, KUDLYAK, AND MEHKARI (2023, *restud*)

- Use ARRA regional variation (2009–2012) at the *county* level.
- Data: household-level retail purchases (Nielsen scanner data) + auto purchases (Equifax).
- **Local consumption multiplier**: $1 of county spending ⇒ $0.29 non-durable + $0.09 autos = 0.38.
- Translate to aggregate with a multi-region NK model with heterogeneous agents, incomplete markets, and trade linkages.
- **Key result**: aggregate consumption multiplier ≈ 0.8 — almost *twice* the local estimate.
  - Trade linkages propagate spending effects across regions.
  - At the ZLB, the aggregate multiplier is even larger.
  - Complete markets model cannot match the positive local multiplier.
- Dupor and Guerrero (2017, *JME*): directly aggregate state-level data to national ⇒ similar but low multipliers (0–0.5).

RAMEY'S CRITIQUE OF SUBNATIONAL ESTIMATES

- Subnational multipliers lie between 1.5 and 2.0 (Chodorow-Reich 2019). But these are *not* the same as national multipliers.
- **Potential upward biases** (Ramey 2019):
  1. *Heterogeneous treatment effects*: if the effect of spending varies across states, unweighted per-capita regressions give equal weight to North Dakota and California. Population weighting lowers estimates substantially (Dupor and Mehkari 2016).
  2. *Spending measure*: federal ARRA spending stimulated additional state and local spending $\Rightarrow$ the federal-only measure understates the total spending increase, biasing the multiplier upward.
- Chodorow-Reich argues the opposite: at the ZLB, subnational estimates are *lower bounds* on the national multiplier (building on Farhi and Werning 2016).
- **Bottom line**: subnational evidence is the best-identified, but translating it to a national number remains contested.

SPENDING COMPOSITION AND INFRASTRUCTURE

DOES THE TYPE OF SPENDING MATTER? (RAMEY 2021)

- Not all government spending is the same. Distinguish between:
  - **Government consumption** G_C: wages, supplies, services.
  - **Government investment** G_I: infrastructure, equipment, R&D.
- Government investment builds public capital K_G which enters production:

$$Y = A K^{\alpha} N^{1-\alpha} K_G^{\theta_G}$$

- Short-run effects may be similar (or investment may even crowd out private investment).
- Long-run effects of investment potentially much larger if $\theta_G > 0$.

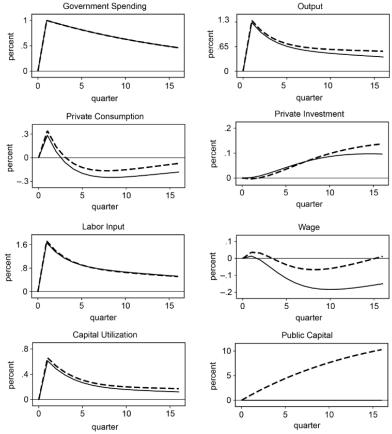


Fig. 4.2 Effect of increases in government consumption or investment, New Keynesian model

Note: Solid government consumption shock; dashed government investment shock, $\theta_G = 0.05$; Goverment spending, output, consumption, private investment, and public capital are expressed in deviations from steady-state value as a percent of output in steady state. Labor input and wages are percentage deviations from their own steady state values. Real interest rate is annualized percentage point deviations from its own steady state.

NK model: government consumption vs. investment IRFs. Ramey (2021)

IMPLEMENTATION DELAYS

- Infrastructure spending faces severe *time-to-build* and *time-to-spend* delays.
- ARRA highway spending (2009): only ≈ 11% spent by June 2009, just over 50% by June 2010, near 100% by June 2012.
- This is devastating for short-run stimulus: by the time infrastructure spending arrives, the recession may be over.
- Government consumption (public employment, supplies) and transfers can be deployed much faster.

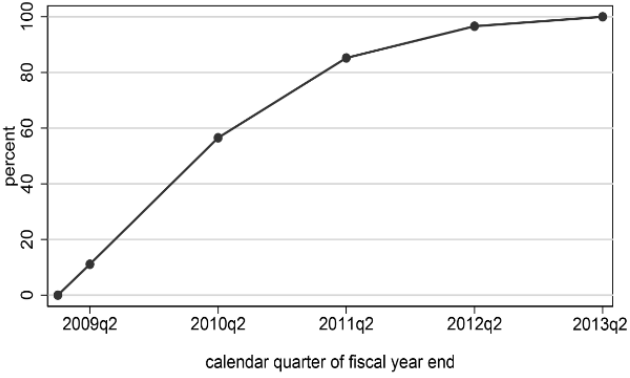


Fig. 4.3 Federal Highway Administration outlays from the American Recovery and Reinvestment Act, cumulative percentage spent of total appropriation

Note: These data are from Leduc and Wilson's (2017) replication files. I aggregated their state-level data to the national level.

ARRA time-to-spend evidence. Ramey (2021)

LONG-RUN INFRASTRUCTURE MULTIPLIERS

- Despite weak short-run stimulus, infrastructure investment can have large *long-run* multipliers.
- With θ_G = 0.05–0.10 (output elasticity of public capital):
  - Present-value long-run multipliers range from 1.3 to 5.4.
  - Infrastructure can potentially *pay for itself* through higher future output.
- Key tension: infrastructure is a poor short-run stabilizer but a potentially powerful long-run growth tool.

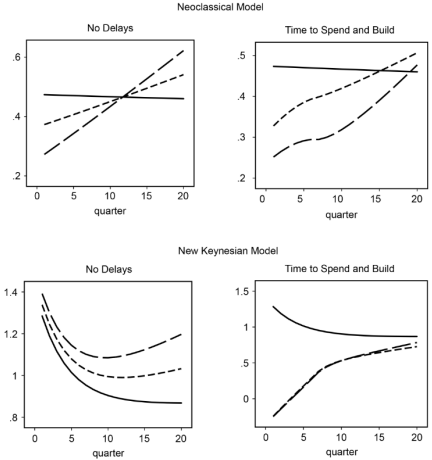


Fig. 4.5 Present discounted value integral multipliers

Note: Solid government consumption shock; short dashed government investment shock, $\theta_g = 0.05$; long dashed government investment, $\theta_g = 0.1$. These estimates are based on the baseline neoclassical and New Keynesian models.

Present-value integral multipliers. Ramey (2021)

Table 4.3 **Long-run multipliers from simulated models**

	Present discounted value		Undiscounted integral	
Model version	Neoclassical	New Keynesian	Neoclassical	New Keynesian
Government consumption	0.44	0.89	0.43	0.90
Initial steady state: government investment/GDP = 3.5%				
No delays				
Government investment, $\theta_G = 0.05$	1.3	1.8	2.4	3.0
Government investment, $\theta_G = 0.10$	2.2	2.8	4.3	5.0
6-quarter time to spend and build				
Government investment, $\theta_G = 0.05$	1.3	1.7	2.4	2.9
Government investment, $\theta_G = 0.10$	2.1	2.5	4.3	4.9
Initial steady-state: government investment/GDP = 1.5%				
No delays				
Government investment, $\theta_G = 0.05$	2.4	3.2	4.9	5.4
Government investment, $\theta_G = 0.10$	4.4	5.4	9.3	9.8
6-quarter time to spend and build				
Government investment, $\theta_G = 0.05$	2.3	2.9	4.9	5.3
Government investment, $\theta_G = 0.10$	4.1	5.0	9.3	9.7

Note: These estimates are based on the calibrated models described in section 4.2. The multipliers are equal to the ratio of the integrals of the impulse responses of output and appropriations. PDV is present discounted value; integral is undiscounted. The top panel shows multipliers from simulations for which the steady-state government investment to GDP ratio is 3.5 percent, which matches the data. The bottom panel shows multipliers from simulations for which the steady-state ratio is 1.5 percent.

Long-run multipliers from simulated models. Ramey (2021)

THE MILITARY MULTIPLIER

THE MILITARY MULTIPLIER (ANTONOVA, LUETTICKE, AND MÜLLER 2025)

- A \$1 increase in military spending does *not* buy \$1 worth of military equipment. Why? Because military buildups bid up the price of military goods.
- Define the **military multiplier** (*MM*): the fraction of a nominal spending increase that translates into additional *real* military equipment.

$$MM = 1 - \frac{\hat{p}_t}{\hat{x}_t}$$

  where $\hat{p}_t$ is the relative price response and $\hat{x}_t$ is the nominal spending increase.
- If prices do not move ($\hat{p}_t = 0$): $MM = 1$ — every dollar buys a full dollar of equipment.
- If prices absorb all the spending ($\hat{p}_t = \hat{x}_t$): $MM = 0$ — pure waste.
- In partial equilibrium: $MM = \left[1 + \frac{1}{\epsilon^d + \epsilon^s}\right]^{-1}$, where ϵ^s, ϵ^d are supply and demand elasticities.
- The *relative price response* is a **sufficient statistic** for *MM*.

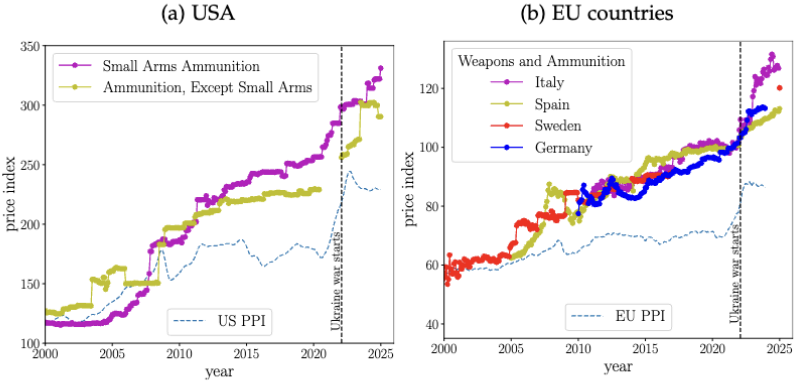
THE MILITARY MULTIPLIER HAS DECLINED

- Antonova et al. (2025) estimate *MM* using local projections with Ramey (2011) military news shocks, splitting into Cold War (1947–1990) and post-Cold War (1991–2018).
- **Cold War**: *MM* ≈ 0.86 at 1 year, rises above 1.0 at 5 years (productivity gains in military production).
- **Post-Cold War**: *MM* ≈ 0.41 at 1 year, only 0.76 at 5 years.
- The price of manufactured goods rises much more strongly and persistently after military news shocks in the post-Cold War period.
- **Why the decline?** The economy's structural shift toward services:
  - Smaller industrial and military sectors ⇒ less capacity to absorb spending increases.
  - Capital reallocation across sectors is costly and time-consuming (Ramey and Shapiro 1998).
  - A multi-sector RBC model with production networks matches the data.

IMPLICATIONS FOR EUROPEAN REARMAMENT

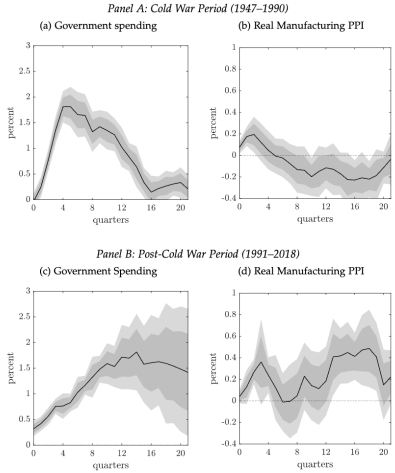
- Since Russia's invasion of Ukraine (2022), arms prices have surged in both the US and EU — consistent with a low *MM*.
- Economic size alone does not guarantee effective military buildups. The EU economy is roughly 10× Russia's, but it may lack the industrial capacity to quickly scale production.
- Key policy levers to raise *MM*:
  - More connected investment and input-output networks (*MM* rises ≈ 50%).
  - Persistent, credible long-term buildup plans (reduces impact multiplier but raises long-run *MM*).
- This complements the fiscal multiplier literature: even if the demand-side multiplier is large, the supply side may limit what military spending actually buys.

Figure 1: Weapon and ammunition prices in US and EU over time



Notes: Sources: USA - BLS PPI data, EU - Eurostat PPI data.

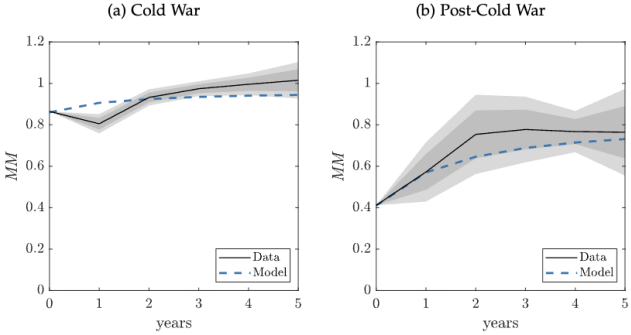
Weapon and ammunition prices in US and EU. Antonova, Luetticke, and Müller (2025)



Notes: Quarterly response of government spending and manufacturing prices (deflated with GDP) to military spending news from Ramey (2016) during the Cold War (1947–1990) and Post-Cold War (1991–2018) periods. We normalize the peak response of government spending to be the same size across samples.

Response to military spending news: Cold War vs. Post-Cold War. Antonova et al. (2025)

Figure 4: Cumulative military multiplier



Notes: Cumulative military multiplier over the first 5 years after the military buildup shock. Left panel: *MM* in Cold War Period (data) and in industrial economy model (large manufacturing sector). Right panel: *MM* in Post-Cold War Period (data) and service economy model (small manufacturing sector). Black solid lines represent the point estimates while shaded areas indicate 68 and 90 percent confidence bounds.

Cumulative military multiplier: Cold War vs. Post-Cold War. Antonova et al. (2025)

LONG-RUN EFFECTS OF GOVERNMENT SPENDING

LONG-RUN EFFECTS REQUIRE LONG DATA (ANTOLIN-DIAZ AND SURICO 2025)

- Standard empirical analyses use post-WW2 data and track responses for 5–20 quarters.
- Antolin-Diaz and Surico (2025): use US quarterly data spanning 1890–2015 with a Bayesian VAR with 60 lags.
- Identification: military spending news shocks (ordered first in Cholesky decomposition).
- Long lag structure + Bayesian shrinkage allow tracking responses up to 15 years.
- Key question: does the multiplier stay constant or change at long horizons?

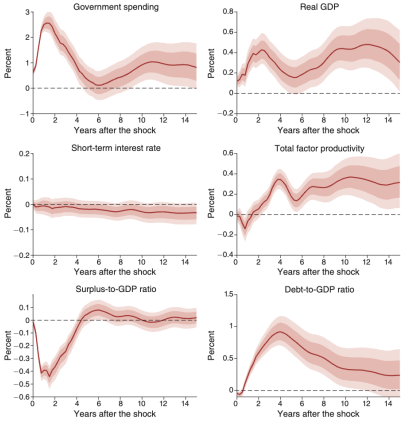


FIGURE 1. IMPULSE RESPONSES TO MILITARY NEWS SHOCK

Notes: The impulse responses are based on an estimated VAR with 60 lags of military spending news, government spending, real per capita GDP, utilization-adjusted TFP, short-term interest rate, fiscal deficit to GDP ratio, and government debt to GDP ratio. Government spending, GDP, and TFP enter the VAR in log-levels. Military spending news is ordered first in the Cholesky factorization. The size of the shock is normalized such as to increase government spending by 1 percent of GDP over the first year after the shock, based on a median G/Y ratio of 19. The darker (lighter) shaded areas represent the central 68 percent (90 percent) high–posterior density (HPD) intervals. The darker solid lines are the median estimates. Results are based on 5,000 posterior draws.

IRFs to military news shock. Antolin-Diaz and Surico (2025)

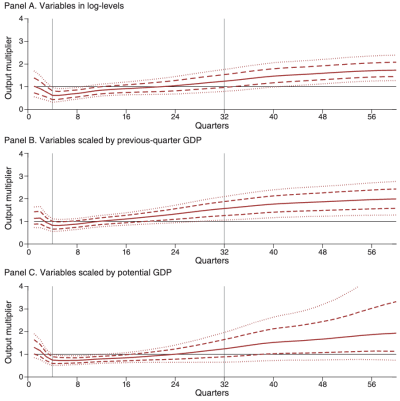


FIGURE 2. THE GOVERNMENT SPENDING MULTIPLIER ACROSS HORIZONS

Notes: The present value multiplier at each horizon h is computed as the ratio of the integral up to horizon h of the output response and the integral up to horizon h of government spending response to a military spending news shock, discounted using the steady-state interest rate. The estimates are based on VARs with 60 lags. In the top panel, government spending and output enter the VAR in log-levels, and the multipliers are obtained using the elasticity formula and the historical median G/Y ratio of 19 percent. In the middle panel, output and government spending are both divided by Y_{t-1}. In the bottom panel, they are expressed in percent of potential output as defined in Ramey and Zubairy (2018). The broken (dotted) lines represent the central 68 percent (90 percent) HPD interval. The solid line stands for the median estimate. Results are based on 5,000 posterior draws.

Government spending multiplier across horizons. Antolin-Diaz and Surico (2025)

WHY DOES THE MULTIPLIER GROW?

- The multiplier rises from ≈ 1 at short horizons to 1.7–2.0 at long horizons.
- This contrasts sharply with standard NK models where the multiplier decays.
- **Mechanism**: military buildups shift the composition of public spending toward R&D.
  - Government R&D expenditure is the *only* component with a large and persistent response.
  - Public investment in E&S responds in the short run, but R&D drives the long run.
- R&D $\Rightarrow$ TFP growth with long diffusion lags (6–17 years) $\Rightarrow$ rising multiplier over time.
- Connects to Ramey (2021): the *composition* of spending matters for long-run effects.

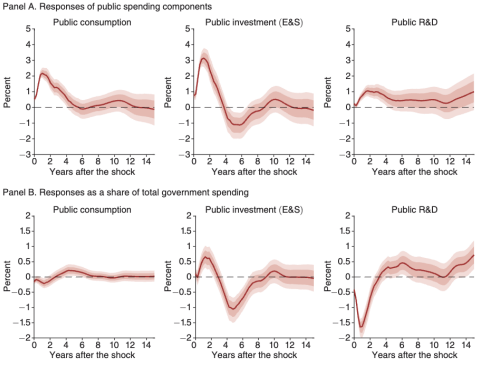


FIGURE 8. EFFECTS OF MILITARY SHOCKS ON PUBLIC SPENDING COMPONENTS

Notes: The impulse responses are based on an estimated VAR with 60 lags of military spending news, real government spending per capita, real GDP per capita, utilization-adjusted TFP, short-term interest rate, fiscal deficit to GDP ratio, and government debt to GDP ratio. In each of the columns, real public consumption per capita, real public non-R&D investment per capita, and real public R&D expenditure per capita, respectively, are added in turn to the VAR. Each government spending category, total government spending and output enter the VAR in log-levels. Military spending news is ordered first in the Cholesky factorization. The size of the shock is normalized such as to increase total government spending by 1 percent of GDP over the first year after the shock. The top (bottom) panel refers to the response of each public spending category in log-level (as share of total government spending). The darker (lighter) shaded area represents the central 68 percent (90 percent) HPD band. The darker solid line stands for the median estimates. Results are based on 5,000 posterior draws.

Effects of military shocks on public spending components. Antolin-Diaz and Surico (2025)

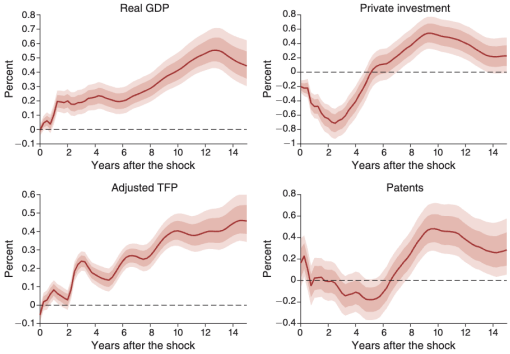


FIGURE 10. IMPULSE RESPONSES TO PUBLIC R&D SHOCK

Notes: The impulse responses are based on an estimated VAR with 60 lags of real public R&D per capita, real total government spending per capita, real GDP per capita, utilization-adjusted TFP, short-term interest rate, fiscal deficit to GDP ratio, government debt to GDP ratio, real private investment per capita, total factor productivity, and patents. Public R&D, total government spending, GDP, and TFP enter the VAR in log-levels. The public R&D shock is identified using the max-share method at the one-year forecast horizon. The size of the shock is normalized such as to increase government spending by 1 percent of GDP over the first year after the shock, based on a median G/Y ratio of 19 percent. The darker (lighter) shaded areas represent the central 68 percent (90 percent) HPD intervals. The darker solid lines are the median estimates. Results are based on 5,000 posterior draws.

Impulse responses to public R&D shock. Antolin-Diaz and Surico (2025)

KEY TAKEAWAYS (I)

1. **Identification matters**: different methods give different short-run dynamics. BP and Ramey news converge on multipliers of **0.6–1.0**; BZP gives higher but less precise estimates.
2. **Methodology matters**: log vs. level specification, multiplier definition, and instrument strength can shift estimates from 0.6 to 2.2.
3. **Subnational evidence**: open economy relative multiplier ≈ 1.5 (Nakamura-Steinsson). Favors NK over neoclassical. But local ≠ aggregate (Dupor et al. 2023; Ramey 2019).
4. **State dependence**: evidence for larger multipliers in recessions is *weaker* than initially thought once methodology is consistent (Ramey-Zubairy 2018).

KEY TAKEAWAYS (II)

5. **Composition matters**: government investment has weak short-run but potentially large long-run multipliers (1.3–5.4); consumption is better for short-run stimulus (Ramey 2021).
6. **Military multiplier has declined**: deindustrialization means each dollar buys less equipment than during the Cold War ($MM \approx 0.4$ vs. 0.9; Antonova et al. 2025).
7. **Long-run multiplier grows**: rises to 1.7–2.0, driven by R&D spillovers to TFP (Antolin-Diaz and Surico 2025).